Course Outline

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I. Introduction

The State of the World and Civilization: Year 2027

Carmichael and Hadikadi, 2019

1. We Live in the Anthropocene
2. State of the Earth: Year 2027
    (a) The Atmosphere
    (b) The Hydrosphere
    (c) The Cryosphere
    (d) The Biosphere
    (e) The Lithosphere
3. State of the Humans: Year 2027
    (a) Current Societal Organizing Principles
    (b) Current Technology
    (c) Current Population
4. State of Civilization: Year 2027
    (a) Globalized
    (b) Weaponized
    (c) Distracted
    (d) Fragile

Opportunities for Planetary Stewardship

1. Stewardship as an Existential Imperative
2. The Critical Earth Systems
   (a) Climate
   (b) Biosphere Productivity
   (c) Biosphere Waste Processing
3. The Critical Human Systems
   (a) Economy
   (b) Governance
   (c) Culture
   (d) Existential
4. External Opportunities for Stewardship
   (a) Creating New Systems
   (b) Re-vamping Existing Systems
5. Internal Opportunities for Stewardship
   (a) Creativity
   (b) Intuition
   (c) Mind, Body, Spirit

II. Approaches

Systems Analysis

1. Fundamental Systems Concepts and Perspectives
   (a) System Components and Characterizations
   (b) Systems Science
   (c) Mental Models

Scale Analysis

1. Spatial Scales
   (a) Global

   (b) Regional
   (c) Local

2. Temporal Scales
   (a) Past
   (b) Present
   (c) Future

Data Analysis

1. Probability Models
2. Statistical Models
3. Agent-Based Models
4. Time Series Models

III. Topics

Concepts and Frameworks

1. Sustainability Concepts
   (a) Brundtland Commission
   (b) UN Sustainable Development Goals
   (c) Marshall and Toffel

2. Sustainability Frameworks
   (a) Ecological Sustainability
   (b) Social Sustainability
   (c) Economic Sustainability
   (d) Agricultural Sustainability

A Sustainable and Resilient Biosphere

1. Functional Terrestrial and Aquatic Ecosystems
   (a) Necessary Inputs
   (b) Necessary Outputs

A Sustainable and Resilient Civilization

1. Culture
   (a) Pilars of Belief
   (b) Open Cooperative Competition
2. Power and Status
   (a) Governance
   (b) Elites
3. Money
   (a) Value vs Money vs Wealth
   (b) Economy and Trade
4. Consumption
   (a) Extraction
   (b) Wastes

References

Carmichael, T., & Hadikadi, M. (2019). The fundamentals of complex adaptive systems. In *Complex adaptive systems: Views from the physical, natural, and social sciences* (pp. 1–16). Springer. (Cit. on p. 1).